# Next-Gen HiRes PureSpikeFormer Architecture

Label-Free Dense Observable Foundation Distillation, Energy-Compensated Masked Event Completion & Pure-Spike Readout

Training-Only (Discarded) · Deployed Event Student · {0, 1} Spikes · Distill Losses

## STAGE A · Training-Only Dense Observable Teacher & Target Bank (Label-Free U1 · Discarded at Inference)

**Paired Train Stream**
(train_ids only · no labels)

**Paired RGB Frame**
$I_i \in \mathbb{R}^{3\times H\times W}$

**Aligned ON/OFF Events**
$X_i \in \mathbb{R}^{16\times 2\times 48^2}$

**FROZEN**

**Pretrained DINOv2 (ViT-S/14)**
Extracts multi-layer RGB CLS & dense 16×16 patch tokens

**Late-Block Weight Init**
Blocks [3, 6, 9, 11] → Student [1, 2, 3, 4]

**Observable Bridge & EventProxy**
Aligns RGB semantics with DVS event stream; filters static unobservable color/texture & retains motion-supported object structure.

**Label-Free Transport**

**Train-Only Dense Target Cache**

**Global Transported Target & Full Bank**
$z_l^{obs} \in \mathbb{R}^{384}$, $B_{tr} \in \mathbb{R}^{N_{tr}\times 384}$, $c_{11} \in \mathbb{R}^{384}$

**HiRes Dense Patch Target (1-to-1)**
$P_{12} \in \mathbb{R}^{16\times 16\times 384}$ (Block 12 tokens)

**Mid-Layer Structural Patch Target**
$P_{10} \in \mathbb{R}^{12\times 12\times 384}$ (Block 10 tokens)

**Multi-Granularity Label-Free Distillation**

$\mathcal{L}_{global} = 2.2\,\mathcal{L}_{align} + \mathcal{L}_{geom} + 0.7\,\mathcal{L}_{nbr} + 0.3\,\mathcal{L}_{mid}$
Cosine align + relational Gram + dual-temp bank KL

$\mathcal{L}_{patch\text{-}aff}$: **Centered Saliency Patch Affinity**
Spatially centered $16^2$ & $12^2$ Gram weighted by $w_{sal}w_{sal}^{\top}$

$\mathcal{L}_{inv}$: **Masked–Clean View Invariance**
$1-\cos(e_{mask}, \mathrm{sg}(e_{clean}))$ on readout & core spikes

DINOv2 Blocks [3, 6, 9, 11] Weight Init

Dense Tokens $(M, S^{(3,4)})$ Supervision

## STAGE B · Deployed Event-Only Student: HiRes PureSpikeFormer (100% Binary {0, 1} Inter-Block Spikes)

**Event Input & Views**

**Clean Event Stream**
$X \in \mathbb{R}^{T\times 2\times 48\times 48}$
T = 16, ON/OFF polarity

**Train View 2: Energy-Masked**
- Shift ±3 px (p_flip = 0)
- 3×3 block drop (18%)
- **L1 Scale [1.0, 1.35]**

preserves BNTT rate!

**Local Spiking Stem**
(patch_size = 3 · per step t)

**3×3 Local Conv (2 → 64)**
BNTT (×16) → PLIF {0, 1}

**3×3 Stride-3 Patch Conv**
BNTT (×16) + 2D PosBias
**Patch PLIF → Binary Spikes**

**Spike Tokens**
1, 0, 1, …, 0, 1
16×16 = 256 (D = 384)

**4 × PureSpikeFormer Blocks**
(D = 384, Heads = 6, T = 16)

- **Block 1** ← DINOv2 Block 3
- **Block 2** ← DINOv2 Block 6 [Readout]
- **Block 3** ← DINOv2 Block 9 [Readout]
- **Block 4** ← DINOv2 Block 11 [Readout]

Global Binary QKV Attn + Local Mixer (d=1,2)
All inter-block tensors strictly ∈ {0, 1}

**Multi-Depth Signed Readout**

**Softmax Fusion (Blocks 2, 3, 4)**
$\Sigma_{t=2}^{4} \alpha_t W_t S_t^{(t)} \rightarrow \mathrm{BNTT}_{2D}$

**Dual-Channel Semantic PLIF**
$s_t^{sem} = s_t^{+} - s_t^{-} \in \{-1, 0, +1\}$

**Learned Temporal Pool** $\Sigma_{t=1}^{16} w_t s_t^{sem}$

**Dense Semantic Map**
$M \in \mathbb{R}^{16\times 16\times 384}$
Sharp object & part tokens (DINOv3 Fig.13)

**Global Event Embedding**
$e \in \mathbb{R}^{384}$ (Saliency + 2×2 Quad)
$e_{blend} = \mathrm{Norm}(e_{unif} + \lambda\, e_{sal})$

**N-Caltech101 U1 kNN: 63.7% / 64.9%**

Detached Event-Saliency Weights $w_{sal} = \mathrm{Pool}_{p\times p}(X_{sum}) + \varepsilon$

## ZOOM-IN · Single PureSpikeFormer Block Circuit (Parallel Global Spike Attention + Local Structure Mixer)

**Input Spikes**
$S_t^{(\ell-1)} \in \{0,1\}^{N\times D}$

**Global Branch: Normalized Binary Spike Self-Attention**
Linear+BNTT+PLIF → Binary $(QK^{\top}/N)V$ → Proj + PLIF

**Local Branch: Event-Observable Structure Mixer**
3×3 Dilated Convs (d = 1, d = 2) + PLIF → Novelty Route $|\Delta S_t|$ Gate × Softmax Mix → $\sigma(\gamma)\cdot$ PLIF

**Residual Re-Spike**
BNTT + PLIF → {0, 1}

**Spiking MLP**
FC1→BNTT→PLIF
FC2→BNTT+Res→PLIF

**Output Spikes**
$S_t^{(t)} \in \{0,1\}^{N\times D}$

## KEY MECHANISMS · Energy Masking & Clean-Train BNTT Calibration

**1. Firing-Rate-Preserving Block Masking**
$X_{mask} = (M \odot X)\cdot \mathrm{clamp}(\|X\|_1/\|M\odot X\|_1,\ 1.0,\ 1.35)$
Paired [X_clean, X_mask] batch prevents BNTT threshold drift.

**2. Student EMA + Population BNTT Recalibration**
EMA ($\mu = 0.995$) + exact clean-train BNTT stats (no_grad).
Zero class labels in student training · zero test-split access.